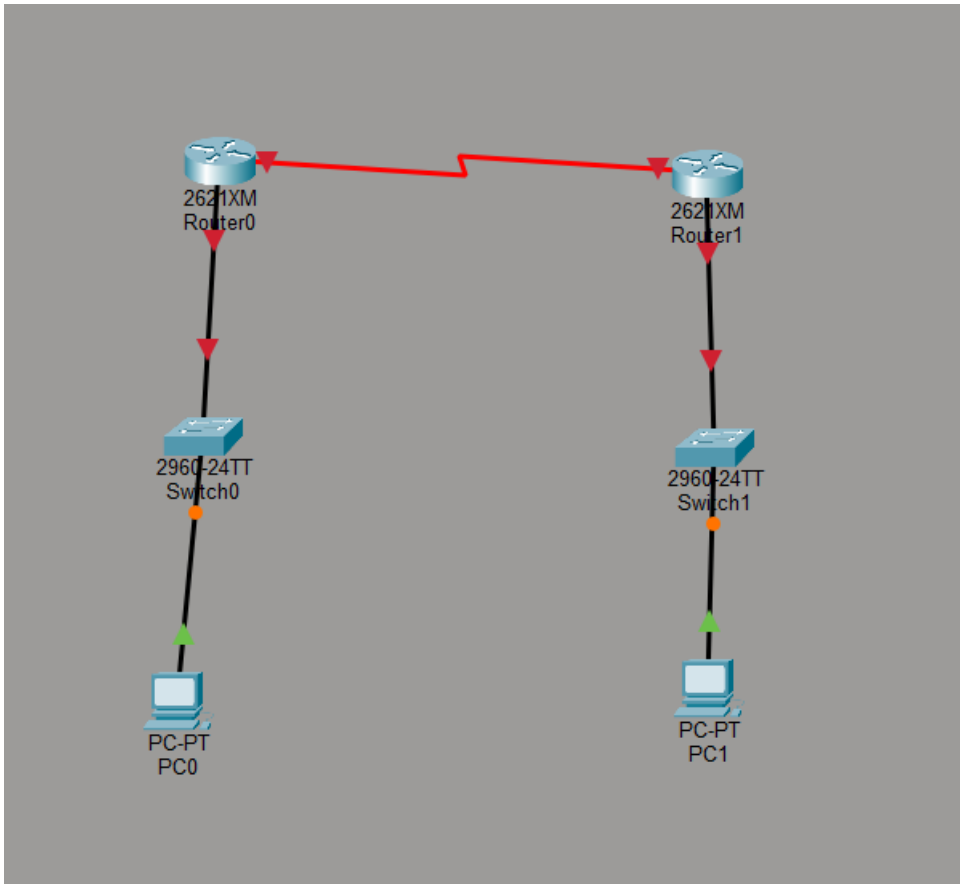


Q-Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.

- Create a simple routed network in Cisco Packet Tracer using two Cisco 2621XM routers connected by a serial link.
- Assign IPv4 addresses to the router LAN and serial interfaces.
- Configure static routes so the two LANs can communicate through the routers.
- Configure PC default gateways and verify connectivity using ping and show ip interface brief.

TOPOLOGY



3. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet	192.168.1.10	255.255.255.0	192.168.1.1
Router0	FastEthernet0/0	192.168.1.1	255.255.255.0	-
Router0	Serial0/0	10.0.0.1	255.255.255.252	-
Router1	Serial0/0	10.0.0.2	255.255.255.252	-
Router1	FastEthernet0/0	192.168.2.1	255.255.255.0	-
PC1	FastEthernet	192.168.2.10	255.255.255.0	192.168.2.1

Router0



Press RETURN to get started!

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief


| Interface       | IP-Address  | OK? | Method | Status                | Protocol |
|-----------------|-------------|-----|--------|-----------------------|----------|
| FastEthernet0/0 | 192.168.1.1 | YES | manual | up                    | up       |
| FastEthernet0/1 | unassigned  | YES | unset  | administratively down | down     |
| Serial0/0       | 10.0.0.1    | YES | manual | down                  | down     |


Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/7/11 ms

Router#
```

Router1

Router1

Physical

Config

CLI

Attributes

Router>enable

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 192.168.2.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#ip address 192.168.1.0 255.255.255.0 10.0.0.1

Router(config)#

% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0	unassigned	YES	unset	administratively down	down

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0	unassigned	YES	unset	administratively down	down

Router#enable

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface serial 0/0

Router(config-if)#ip address 10.0.0.2 255.255.255.252

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface Serial0/0, changed state to up

Router(config-if)#exit

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0	10.0.0.2	YES	manual	up	up

Router#ping 10.0.0.1

PC0

PC0

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.1.10

Subnet Mask

255.255.255.0

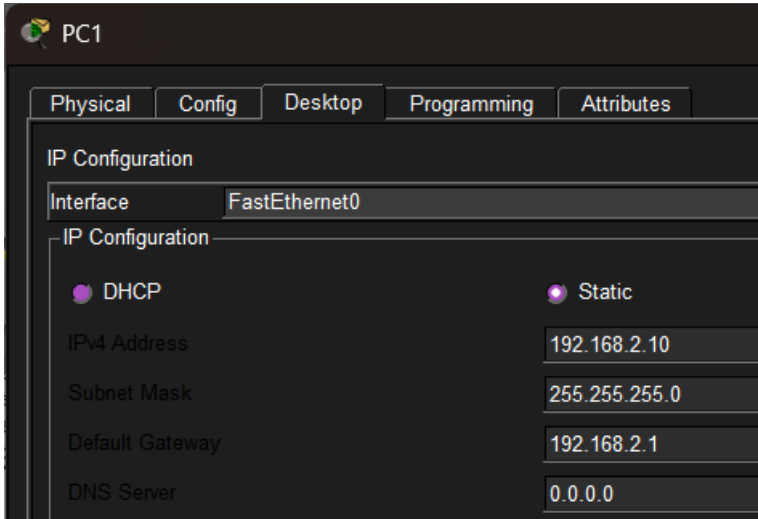
Default Gateway

192.168.1.1

DNS Server

0.0.0.0

PC1



TESTING

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=12ms TTL=126
Reply from 192.168.2.10: bytes=32 time=5ms TTL=126
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 5ms

C:\>
```